AI -POWERED DATA ANONYMIZATION FOR DPEARTMENT OF CENSUS AND STATISTICS OF SRI LANKA

25-26J-123

Project Proposal Report

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Evaluating Re-identification Risks in Anonymized Data Using Machine Learning

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Declaration

I say that this is my own work, and this project does not use any work that was given before for a degree or diploma at any other university or college without saying where it came from. As far as I know, it does not have any work that was published or written by another person before, except where I have said who wrote it in the text.

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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

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(Dr. Prasanna Sumathipala)

Date.....2025/02/18

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Abstract

AI-driven framework for **evaluating re-identification risks in anonymized data using machine learning** within the context of datasets from the Sri Lankan Department of Census and Statistics. This project focuses on solving the problem of balancing data utility with individual privacy. Government agencies need to release data for research, but at the same time, they must make sure that no person can be identified. Traditional anonymization methods, like suppression or generalization, are not enough, because modern attackers can still re-identify individuals by using advanced techniques. Static measures also fail to capture the real complexity of machine learning–based linking attacks. This situation creates a clear research gap that our work aims to fill.

Our method uses a step-by-step approach. First, we simulate an attacker's behavior with supervised machine learning models such as Random Forest and Gradient Boosting. These ensemble models are chosen because they are powerful, accurate, and reliable when handling complex patterns in the data. By training these models to perform linking attacks, we can measure their success rate, which then serves as a dynamic and realistic re-identification risk score. This gives a more practical view of how vulnerable anonymized data really is.

A major innovation of this project is the use of Explainable AI (XAI). By applying SHAP (SHapley Additive exPlanations) values, the system does not only produce a single risk score but also shows which attributes, like age, location, or occupation, contribute the most to the risk. This makes the results easier to understand and more actionable. Policymakers and data custodians can clearly see where the weaknesses are and take steps to improve anonymization.

The final outcome of the project will include overall risk metrics, attribute-level vulnerability analysis, SHAP-based visualizations, and adaptive recommendations. This framework is both dynamic and context-specific, providing a valuable solution for Sri Lankan census data privacy.

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List of Abbreviations

AI	Artificial Intelligence
ML	Machine Learning
XAI	Explainable Artificial Intelligence
SHAP	SHapley Additive exPlanations

1.0 INTRODUCTION

This research addresses the problem of **re-identification risk in anonymized data**, especially in Sri Lankan census datasets. Governments and institutions share data for transparency and research, but even anonymized records can often be linked back to individuals. Past studies introduced methods like k-anonymity and l-diversity, but these are static and cannot stop modern **machine learning–driven attacks**.

The literature shows that while many **Privacy-Preserving Data Publishing (PPDP) techniques** exist (e.g., probabilistic k-anonymity, location-based anonymization, GAN-based methods like AnomiGAN), they still face the trade-off between **privacy and utility**. However, most global studies lack **interpretability** and do not apply to **Sri Lankan datasets**.

The research gap is therefore threefold: (1) absence of **dynamic ML-based risk assessment**, (2) limited **explainable insights**, and (3) lack of **Sri Lanka–specific analysis**.

The research problem is clear: the Department of Census and Statistics must release useful datasets but also **protect privacy**. Traditional anonymization is no longer sufficient, and there is a need for a **framework that is dynamic, interpretable, and tailored to Sri Lankan data**.

1.1 Background and Literature

This research proposal addresses a pressing issue in data privacy: the re-identification risk inherent in anonymized datasets. In the age of big data and open science, government agencies and research institutions are increasingly publishing large datasets to foster transparency and innovation. However, this practice creates a significant challenge, as even seemingly anonymous data can be linked with external information to re-identify individuals, thereby compromising privacy. This project proposes a novel, data-driven framework to dynamically assess this risk, specifically tailored to the context of Sri Lankan census data.

The literature review will focus on three key areas to provide a comprehensive overview of the current research landscape:

Re-identification Attacks on Anonymized Data: More recent studies have evolved Sweeney's foundational work by using advanced machine learning models to automate and enhance linking attacks, demonstrating the persistent vulnerability of anonymized data. This evolution marks a shift from theoretical attacks to practical, data-driven threats, necessitating more robust defense mechanisms [1] [2], [9]

Privacy-Preserving Data Publishing (PPDP) Techniques: This field has developed various methods to mitigate re-identification risk. Approaches such as probabilistic k-anonymity [2], location-based anonymization [3], GAN-based anonymization (AnomiGAN) [4], utility-aware anonymization [5], and time-series anonymization [6] attempt to balance privacy with data utility. Frameworks such as X2R2 [1] provide explainable risk exploration, and libraries like Anjana [7] offer open-source anonymization toolkits. Broader surveys integrating technical and legal perspectives emphasize this trade-off between data sharing and protection [8], [9].

Explainable AI (XAI) for Privacy and Security: The application of XAI in privacy is an emerging and critical area. Techniques such as SHAP values [10] enable interpretable explanations for complex ML models by quantifying feature contributions. Recent surveys provide taxonomies of explainability techniques across domains, including privacy-sensitive data analysis [11]. Frameworks such as X2R2 [1] apply XAI directly for re-identification risk analysis, while Simon et al. [9] propose structured approaches for assessing and mitigating risk in healthcare data. Together, these contributions highlight the importance of combining dynamic ML-based risk assessment with transparent explanations to strengthen trust and inform policymaking.

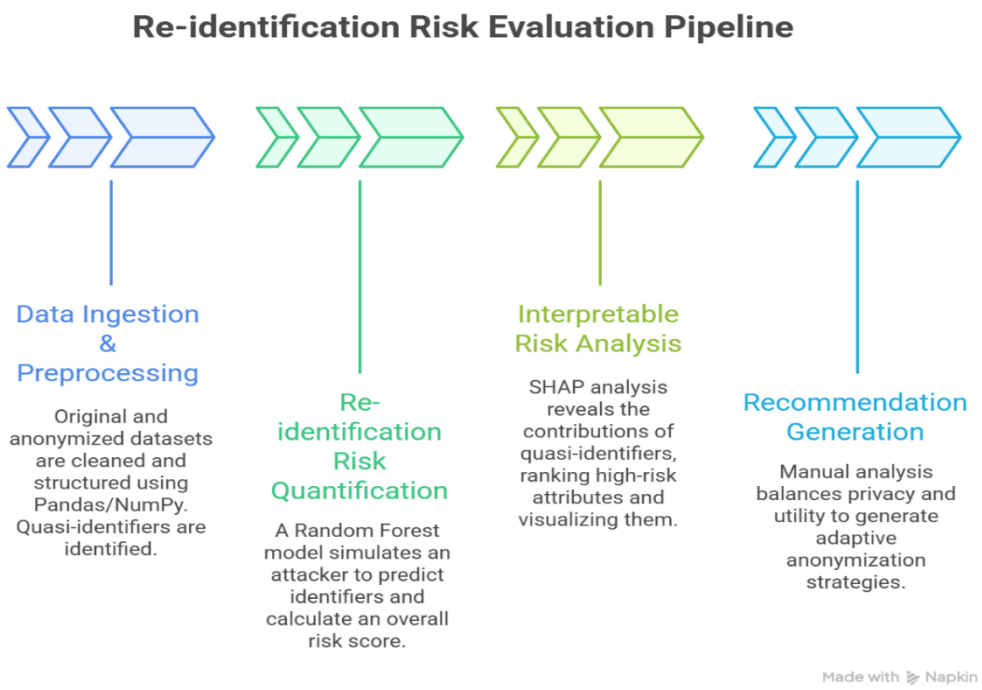

Figure 1:Pipeline for Privacy Risk Analysis

1.2 Research Gap

Lack of a Dynamic, ML-based Risk Assessment: Most existing studies use static measures such as uniqueness counts or k-anonymity thresholds [1]– [4], [6]. These simple rules cannot capture the full complexity of modern attacks, where machine learning can exploit hidden patterns in data. There is a gap for a framework that directly simulates an attacker using advanced ML techniques to give a more realistic and dynamic risk assessment.

Limited Interpretability in Risk Analysis: Many current methods only produce a single risk score, without showing which attributes or combinations of attributes cause the risk [9]– [11]. This lack of explanation makes it hard for data custodians to decide which attributes need stronger anonymization. Adding Explainable AI (XAI) would provide transparency, highlighting which features contribute most to re-identification risk and supporting better decisions.

Context-Specific Analysis for Sri Lankan Datasets: Global research on re-identification is available, but very little work has been done on Sri Lankan census data [2], [8], [9]. Privacy risks depend on cultural, social, and local data factors. An attribute that is safe in one country may be highly identifying in another. Because of this, there is a clear need for a framework tailored to Sri Lankan datasets, so the Department of Census and Statistics can make context-specific decisions.

The literature shows three major gaps: (1) a lack of dynamic ML-based risk assessment, (2) limited interpretability in existing methods, and (3) the absence of Sri Lanka–specific studies. These gaps call for a new approach that is dynamic, interpretable, and context aware.

1.3 Research Problem

In an era of increasing data-driven governance and open data initiatives, the Sri Lankan Department of Census and Statistics faces a critical challenge: publishing valuable, large-scale datasets for research and public use while simultaneously safeguarding individual privacy. The core research problem lies in the inherent tension between maximizing data utility and minimizing the risk of re-identification.

While traditional data anonymization techniques have been employed to mitigate this risk, they have proven to be insufficient against modern, sophisticated attacks. The problem is threefold:

- **Vulnerability to Advanced Attacks:** Conventional anonymization methods (such as k-anonymity) are static and fail to account for the complex, non-linear relationships between quasi-identifiers that can be exploited by modern machine learning-based linking attacks. A dynamic, data-driven approach is needed to accurately assess re-identification risk in the face of such advanced attack methods [1]– [4].

- **Lack of Interpretability:** Existing risk assessment methods often provide a single, opaque risk score that offers no insight into which specific data attributes are the primary drivers of vulnerability. This lack of transparency makes it difficult for data custodians to make informed, targeted decisions about which columns or features require more stringent anonymization. Explainable AI techniques such as SHAP [10] and XAI taxonomies [11] directly address this issue.

- **Absence of Context-Specific Analysis:** Analysis: While considerable research on data privacy and re-identification exists globally, there is a notable absence of publicly available analysis tailored to Sri Lankan datasets. Privacy threats can vary significantly by context; therefore, a framework specific to the Sri Lankan demographic and data characteristics is essential to produce relevant and actionable insights for local policymakers [2], [8], [9].

This research aims to address these critical issues by developing a framework that not only quantifies the re-identification risk but also provides a transparent, interpretable breakdown

of the vulnerabilities, thereby offering a practical solution for data privacy challenges in the Sri Lankan context.

Paper	Technologies Used	Strengths	Limitations
Hagedoorn et al. (2020) – X^2R^2	ARX-based anonymization, risk metrics	User-friendly, shows privacy–utility tradeoff	Limited to simple risk metrics, not automated
Senavirathne & Torra (2021)	Probabilistic k-anonymity, permutation	Preserves utility, avoids suppression	Distorts correlations, attribute disclosure risk
Shaham et al. (2020)	ML-based clustering + sequence alignment	High utility for trajectory data	Specific to location data, computationally heavy
Bae et al. (2019) – AnomiGAN	GAN + disease classifier	Better utility than DP, balances privacy/accuracy	No formal DP guarantee, resource-intensive
Wang et al. (2018)	Visual anonymization tool (k-anon, DP)	Real-time utility feedback, interactive	Semi-manual, limited scalability
Hashemi et al. (2023)	GAN for time-series	Generates realistic, useful synthetic data	No formal privacy guarantee, heavy training
Díaz & López (2024) – *anjana*	Python library (k-anon, l-div, t-close)	Open-source, covers many models	Lacks evaluation metrics, CLI only
Aufschläger et al. (2023)	Survey + taxonomy (GDPR focus)	Links tech & legal views, clarifies terms	No new method, only tabular data

Table 1 :Technologies, Strengths, and Limitations of Prior Work

2.0 Objectives

2.1 Main Objective

The main objective of this research is to develop and validate a data-driven framework for the Sri Lankan Department of Census and Statistics to evaluate re-identification risks in anonymized data. This framework will use machine learning to simulate linking attacks and employ explainable AI to provide transparent, interpretable, and actionable insights. The ultimate goal is to enable policymakers and data custodians to make informed decisions that effectively balance data utility with individual privacy.

2.2 Specific Objectives

The following are the sub-objectives of conducting this research.

- **Prepare and Preprocess Datasets:** Systematically collect, clean, and preprocess the provided original and anonymized datasets from the Sri Lankan Department of Census and Statistics, identifying key quasi-identifiers to be used in the analysis.
- **Quantify Re-identification Risk:** Develop and train a machine learning model to simulate a re-identification attack, thereby quantifying the overall re-identification risk as a measurable success rate.
- **Provide Explainable Insights:** Integrate explainable AI techniques (such as SHAP values) to identify which attributes contribute most to the re-identification risk, offering an interpretable explanation for the model's risk predictions.
- **Recommend Anonymization Improvements:** Based on the risk analysis, generate actionable recommendations for enhancing the anonymization of the data (e.g., suggesting which attributes to generalize or suppress further) to reduce identified vulnerabilities while preserving as much data utility as possible.

3.0 METHODOLOGY

3.1 Requirement Gathering

This section outlines the essential requirements for the successful execution of this research project, categorizing them into data, software, hardware, and expertise needs. These requirements are critical for implementing the proposed methodology and achieving the research objectives.

3.1.1 Past Research Analysis

A thorough analysis of past research has informed the requirements for this project. The literature review highlighted a critical need for a dynamic, data-driven approach to re-identification risk that goes beyond static metrics. This necessitates software and expertise capable of simulating sophisticated machine learning attacks and utilizing explainable AI techniques. The analysis of existing frameworks, such as X^2R^2 and AnomiGAN, has set a benchmark for the project's functional and technical requirements, emphasizing the need for both accuracy and interpretability in the final solution

Key Requirement / Feature	Other Research	Proposed System
Risk Assessment Method	Static methods like k-anonymity, l-diversity, uniqueness counts .	Dynamic ML models (Random Forest, Gradient Boosting) simulate real attacker behavior.
Interpretability of Results	Provides only a single opaque risk score .	Uses Explainable AI (SHAP) to show which attributes contribute most to risk.
Dataset Focus	Based on US, European, or healthcare datasets .	Tailored to Sri Lankan census data, reflecting local demographic and cultural context.
Anonymization Approaches	Generalization, suppression, masking, GAN-based synthetic data .	Combines traditional anonymization + synthetic data generation + ML-based validation.
Guidance for Policymakers	Limited or missing in most frameworks .	Provides actionable recommendations (e.g., generalize age, suppress rare occupations).

Table 2 :Past Research Analysis

3.1.2 Component Specific System Diagram

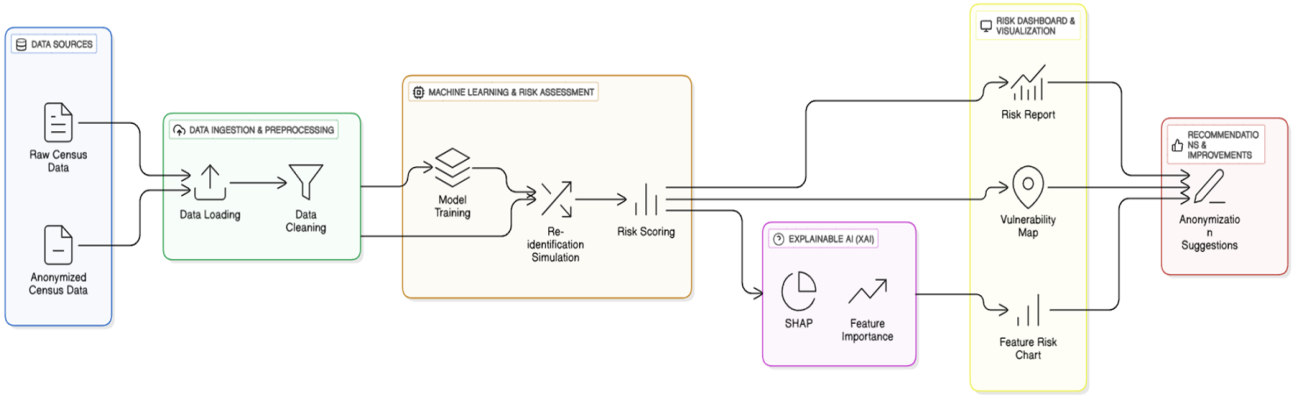

Figure 2:System Diagram

Workflow for Re-identification Risk Assessment

1. Data Sources
    - Raw census data and anonymized census data are used as inputs.
2. Data Ingestion & Preprocessing
    - Data is loaded and cleaned to prepare it for analysis.
3. Machine Learning & Risk Assessment
    - ML models are trained to simulate re-identification attacks.
    - The success rate of these simulations produces a risk score.
4. Explainable AI (XAI)
    - SHAP is applied to highlight feature importance.
    - Shows which attributes contribute most to re-identification risk.
5. Risk Dashboard & Visualization
    - Outputs include a risk report, vulnerability map, and feature risk chart.
6. Recommendations & Improvements
    - Based on results, the system provides anonymization suggestions to reduce risk while maintaining utility.

3.1.3 System Overview

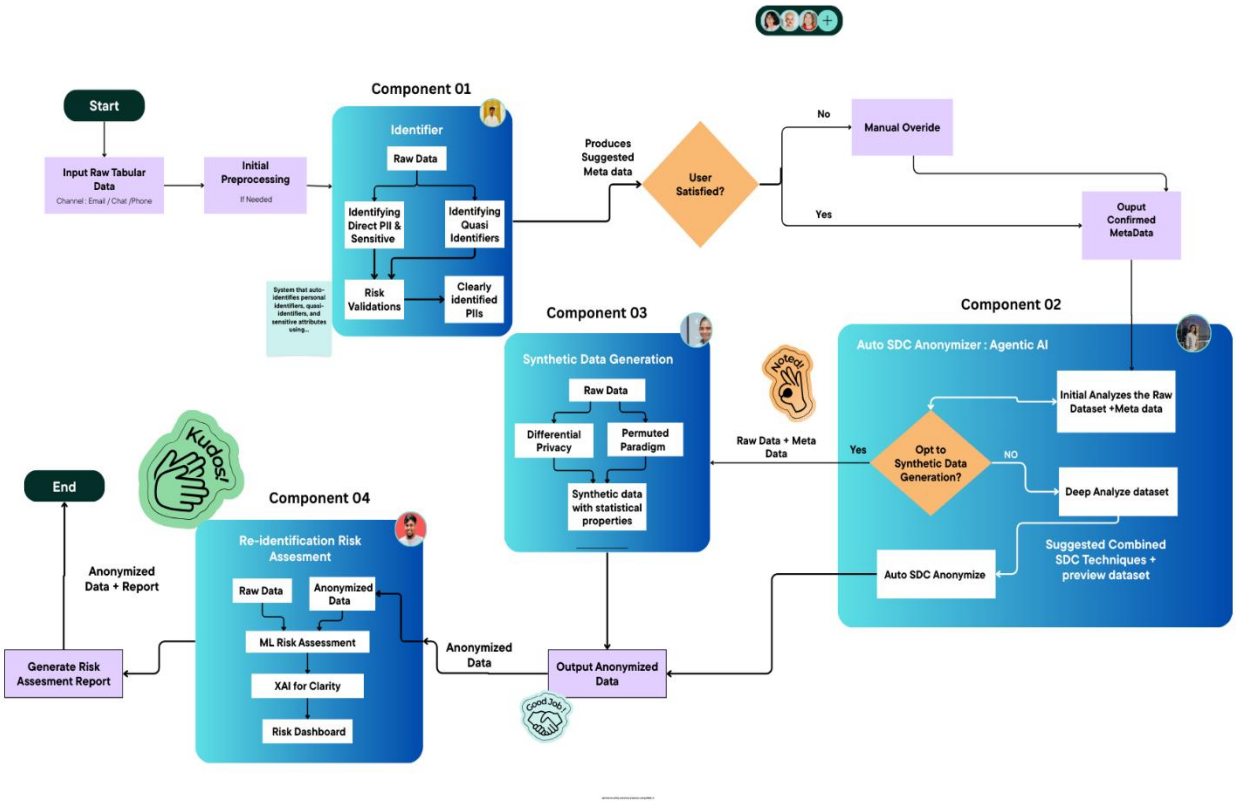

Figure 3: System Overview

Workflow for Data Anonymization

1. Identifier Detection: Find personal identifiers, quasi-identifiers, and sensitive attributes.
2. Anonymization Techniques: Apply methods like generalization or suppression; if needed, move to synthetic data generation.
3. Synthetic Data Generation: Create privacy-preserving synthetic datasets with similar statistical properties.
4. Risk Assessment: Test datasets with ML and XAI for re-identification risk, generate a risk report, and finalize if risk is acceptable.

Summary: Sensitive data is protected through anonymization or synthetic data, with risks checked before release.

3.2 Feasibility Study

3.2.1 Technical Feasibility

3.2.1.1 Knowledge on Technologies

The core technologies required—including machine learning, ensemble methods (Random Forest, Gradient Boosting), and explainable AI (SHAP)—are well-documented and widely supported. The expertise available is sufficient to implement these technologies to address the research problem effectively

3.2.1.2 Knowledge on Tools

The project relies on well-established, open-source tools within the Python ecosystem, such as Pandas, Scikit-learn, XGBoost, and the SHAP library. These tools are mature and have extensive community support, mitigating technical risks.

3.2.1.3 Data collection Knowledge

The project's success is dependent on the availability of the original and anonymized datasets from the Sri Lankan Department of Census and Statistics. As per the project scope, these datasets have been provided, ensuring this critical data collection requirement is met.

3.2.2 Schedule Feasibility

The project scope is manageable and can be completed within the standard academic timeframe for a research proposal. The phased approach, from data preprocessing to final report generation, allows for a structured and realistic schedule.

3.2.3 Economic Feasibility

The project is highly economically feasible. The primary software tools and libraries are open-source and free to use, eliminating licensing costs. The hardware requirements are standard for a research environment, and no specialized or expensive equipment is needed.

3.3 System Analysis

3.3.1 Software Solution Approach

The software solution will be developed as a framework for re-identification risk analysis. This framework will:

1. **Ingest and Preprocess Data:** Efficiently load and preprocess both original and anonymized datasets while identifying and handling key quasi-identifiers.
2. **Simulate Re-identification Attack**: Implement a supervised machine learning model, such as a Random Forest or Gradient Boosting classifier, to simulate sophisticated linking attacks on the anonymized data.
3. **Analyze Risk:** Quantify the overall re-identification risk by evaluating the model's linking success rate and rank quasi-identifiers based on their contribution to this risk.
4. **Explain Results:** Integrate an Explainable AI (XAI) library, specifically SHAP, to provide interpretable and transparent explanations for the risk scores, revealing which data attributes pose the greatest privacy vulnerabilities.
5. **Generate Recommendations**: Provide actionable, adaptive anonymization strategies based on risk analysis to improve data privacy while maintaining data utility

3.3.2 Tools & Technology

- Programming Languages – Python
- NLP & ML – Scikit-Learn, XGBoost, LightGBM
- Database Systems – Google Cloud SQL
- Explainable AI (XAI) - SHAP (SHapley Additive exPlanations)
- Visualization - Matplotlib, Seaborn
- Web Front end : React.js, Tailwind CSS/MUI for UI
- UI Design Implementation – Figma
- Diagramming Tool – Draw.io
- Integrated Development Environment (IDE) – VS Code
- Version Control System (VCS) – Git (GitHub or GitLab)
- Collaboration Tools – Microsoft Teams or WhatsApp.
- Testing Tools - pytest (Python), Postman (APIs), Jest (JS).
- Deployment Tools - Docker, Kubernetes, Jenkins (CI/CD), Ansible/Chef (config).
- Code Quality - SonarQube.
- Project Management- MS Planner, Jira

3.4 Project Requirements

3.4.1 Functional Requirements

The functional requirements for the proposed model are as follows:

- **Data Ingestion and Preprocessing:**
  The system must be able to load both original and anonymized census datasets, clean the data, and identify key quasi-identifiers such as age, gender, and location.

- **Machine Learning Attack Simulation:**
  The system must train supervised ML models (e.g., Random Forest, Gradient Boosting) to simulate re-identification attacks on the anonymized data.

- **Risk Score Calculation:**
  The framework must output an overall re-identification risk score based on the success rate of the ML models.

- **Attribute-Level Risk Analysis:**
  The system must identify and rank the quasi-identifiers by how much they contribute to re-identification risk.

- **Explainable AI (XAI) Insights:**
  The framework must integrate SHAP to provide interpretable explanations, showing which attributes or combinations of attributes cause the highest risk.

- **Visualization:**
  The system must generate clear visualizations (charts, SHAP plots) so that results are easy to understand for policymakers and data custodians.

- **Adaptive Anonymization Suggestions:**
  Based on the analysis, the framework must generate practical recommendations (e.g., generalize age groups, suppress rare occupations) to improve data privacy while keeping data useful.

3.4.2 Non-Functional Requirements

The non-functional needs that were prioritized throughout the creation of the suggested model are listed below.

- **Usability**: The framework should have a simple and clear interface so that even users without advanced technical knowledge (e.g., policymakers or staff from the Department of Census) can use it easily with minimal training.

- **Reliability**: The system should consistently provide correct results without crashing or unexpected failures during tasks like dataset loading, risk analysis, or SHAP visualizations.

- **Availability**: The system should be available for authorized users (such as Census Department staff or researchers) whenever needed during working hours, either on local servers or through a secure online platform.

- **Accuracy**: The risk scores and explanations should be based on properly trained and validated ML models, ensuring that results are trustworthy and reflect real re-identification risks.

- **Performance**: The system should analyze datasets and generate results (risk scores, SHAP visualizations) within a reasonable time, avoiding long delays even for large census datasets.

- **Security**: Sensitive census data must be fully protected. The system should prevent unauthorized access, ensure secure storage, and avoid any leakage of private information during analysis or reporting.

3.5 Testing

A robust testing strategy will be implemented to ensure the framework's accuracy and reliability. The testing will include:

- **Unit Testing:** Test individual components such as data preprocessing, machine learning model training, risk score calculations, and Explainable AI (SHAP) computations to ensure each function works correctly.
- **Integration Testing:** Verify proper data flow and interaction between components from data ingestion to preprocessing, model training, risk scoring, and explanation generation—to ensure they work seamlessly together.

- **Validation Testing:** Validate overall outputs against known privacy principles and controlled test cases to ensure risk scores are accurate, logical, and that explanations correctly identify key risk factors.
- **Security Testing:** Make sure no sensitive data leaks occur during testing, safeguarding confidential information throughout the process.

3.6 Timeline

The proposed timeline for the project is as follows.

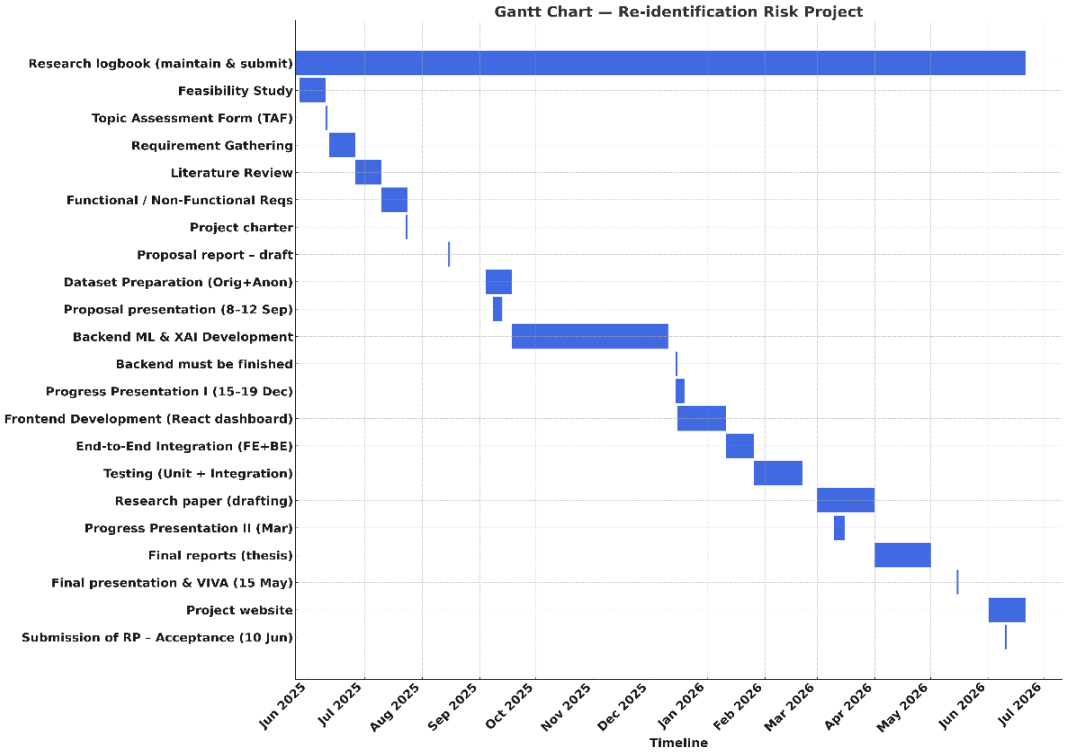

Figure 4: Gantt Chart for Project Schedule Management

3.7 Communication Management Plan

This plan defines who needs what information, how it's shared, when it's shared, and who is responsible—so the supervisor, co-supervisor, team, and external advisors can make timely decisions and keep the project on track.

3.7.1 Communication Objectives:

Proactive communication is important on all projects. Communication needs to be:

- Adequate: in the right format and right content.
- Specific: for the targeted audience.
- Sufficient: providing all the necessary information.
- Concise: brief, avoiding repetition and non-important information.
- Timely: addressing points at the right time.

3.7.2 Communication Media:

The communication media that will be used for the project are:

- Primary: Microsoft Teams (meetings/chat), Email (formal notices/approvals).
- Work tracking: Click Up/Trello (tasks), GitHub (code, Issues, PRs, Wiki).
- Quick updates: WhatsApp (urgent coordination only).
- Documents: Google Drive/OneDrive folders with version history.
- Meeting minutes: shared Doc (dated; action items with owners & due dates).

Audience	Content	Format/Channel	Frequency	Owner	Success Criteria
Supervisor & Co-supervisor	Progress, risks, decisions needed, upcoming milestones, and overall findings	Slide deck + email; MS Teams meeting	Bi-weekly	Project Lead	Actions are agreed upon; feedback is integrated into the project plan.
Project Team	Tasks, blockers, data updates, and code integration issues.	Click Up/Trello updates; GitHub Issues/PRs; Teams chat	Weekly + as needed	Module Owners	All critical tasks are completed on time; blockers are resolved efficiently.
Department of Census & Statistics	Project status, risk analysis findings, re-identification risk scores, and final recommendations.	Formal presentation; Brief memo + email	Monthly / per milestone	Project Manager	Findings and recommendations are clearly understood and accepted by stakeholders.

Table 3: Communication Management Plan

4.0 COMMERCIALIZATION

The commercialization of this re-identification risk evaluation framework is aimed at delivering practical and strategic value to the Department of Census and Statistics (DCS). The Department is responsible for releasing large amounts of census and survey data for research, planning, and policymaking. By using this framework, the DCS can ensure that sensitive information is better protected while still allowing the data to be useful for analysis. This strengthens public trust and supports evidence-based decision-making in Sri Lanka.

Unlike traditional methods that depend on fixed anonymization rules, this framework uses machine learning to simulate re-identification attacks and Explainable AI (XAI) to identify which attributes create the most risk. This means the Department will not only know the overall level of risk in a dataset but will also see which fields, such as age or location, require stronger anonymization. This gives the DCS practical guidance before publishing anonymized datasets.

The implementation strategy is phased. First, the framework will be delivered as an open-source tool that can run within the Department's IT environment. Second, training sessions will be given to DCS staff so they understand how to use the tool and interpret the risk reports. Third, the research team will provide technical support during the adoption stage, helping the Department adjust anonymization strategies based on the results.

Because the tool will be open source, it ensures sustainability. The Department can continue updating and extending the system over time, applying it not only to census data but also to other sensitive datasets such as household surveys, education, and health statistics.

5.0 BUDGET

The budget for this project is designed to be lean and efficient, leveraging open-source technologies and existing resources to minimize costs. The primary expenses are related to personnel and indirect project-related activities, with a total estimated cost for the entire research period However, there will be some other costs expected to as be given in the table below.

Category	**Cost (LKR)**	**Justification**
Computing (Cloud / Contingency)	10,000	For running ML models, XAI visualizations, and possible cloud-based storage/VMs if local resources are insufficient.
Stationery & Printing	5,000	Printing draft reports, final thesis, diagrams, charts, and other project documentation.
Software & Libraries	0	All core tools (Python, Scikit-learn, XGBoost, SHAP, React.js, Docker, GitHub) are open-source.
Development Tools / IDE	2,000	Small allowance for licensed plugins, diagramming tools (e.g., Draw.io premium / Figma elements).
Internet & Communication	6,000	High-speed internet for dataset transfer, virtual meetings (MS Teams/Zoom), GitHub access, cloud uploads.
Workshops / Expert Consultation	8,000	Possible expenses for consulting with domain experts (Sri Lanka Census Dept.) or attending seminars.
Miscellaneous / Contingency	4,000	Buffer for unforeseen costs (transportation, backup storage devices, small hardware replacements).
Total	35000	

Table 4:Cost Management Plan

This amount may differ according to the economic crisis.

6.0 SUMMARY

This research proposal presents a comprehensive plan to develop a machine learning and explainable AI (XAI)–driven framework for evaluating re-identification risks in anonymized data. The Department of Census and Statistics faces the critical challenge of maximizing data utility while minimizing privacy risks. Existing methods often rely on static metrics and provide limited interpretability, creating a gap that this project aims to address.

The proposed methodology simulates adversarial behavior using advanced machine learning techniques to quantify re-identification risk dynamically. Integrated XAI methods, such as SHAP, will provide transparent explanations of which quasi-identifiers contribute most to vulnerabilities. This dual approach ensures that the framework answers not only "how much risk" exists but also "why" and "how" it can be mitigated.

Key components include data ingestion, preprocessing, risk quantification, interpretability analysis, and anonymization recommendations. Feasibility is ensured through open-source tools, established libraries, and the research team's expertise. A structured timeline, risk management plan, and communication strategy are in place to ensure progress and stakeholder engagement.

The project's outcomes will deliver a validated tool and practical guidelines, enabling the Department to share anonymized data safely. Ultimately, this framework will strengthen trust, improve confidence in data sharing, and contribute to global privacy-preserving data practices

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